

Rossmoyne Senior High School

Semester One Examination, 2018

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section Two:
Calculator-assumed

SOLUTIONS

Student number: In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	52	35
Section Two: Calculator-assumed	14	14	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet.
3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.
4. Supplementary pages for the use of planning/continuing your answer to a question have been provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
6. It is recommended that you do not use pencil, except in diagrams.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed

65% (98 Marks)

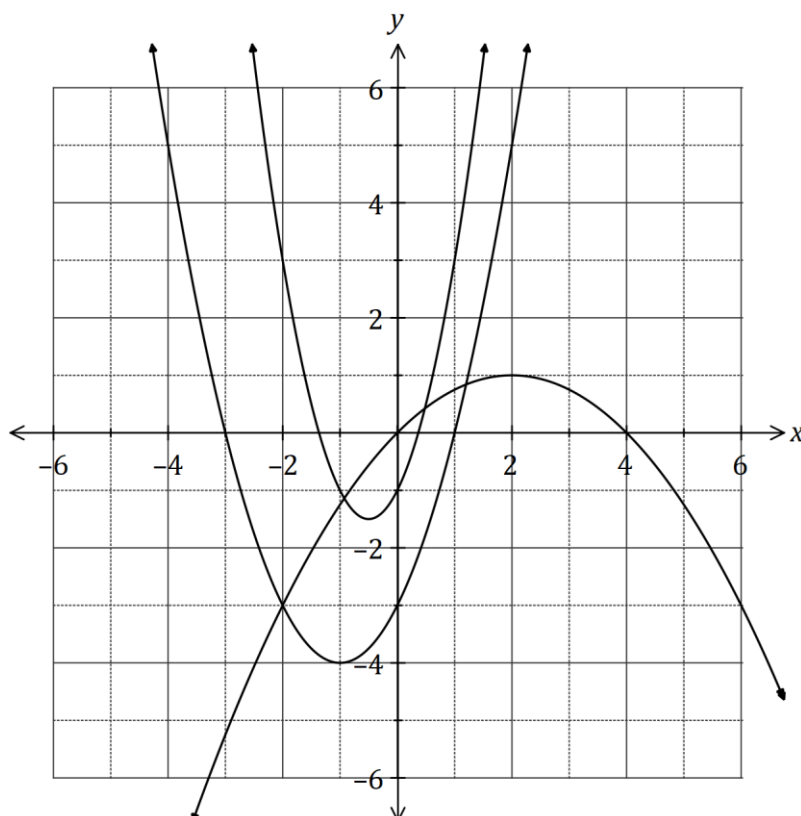
This section has **thirteen (13)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 9

(3 marks)

The graphs of $y = 2x^2 + 2x + c$, $y = a(x - 2)^2 + 1$ and $y = (x + b)(x + 3)$ are shown below.



Determine the values of the constants a , b and c .

Solution
$x = 4 \Rightarrow 0 = a(4 - 2)^2 + 1$ $a = -\frac{1}{4}$
$b = -1 \text{ (Other root at -3)}$
$c = -1 \text{ (y-intercept)}$
Specific behaviours
✓ value of a ✓ value of b ✓ value of c

Question 10**(7 marks)**

An online store is offering new customers the opportunity to select 5 different products for just \$5 each. They can select from a range of 8 different books, 12 different magazines and 16 different games.

Determine how many different selections can be made if:

- (a) there are no restrictions.

(2 marks)

Solution
$8 + 12 + 16 = 36$ ${}^{36}C_5 = 376\,992$
Specific behaviours
✓ indicates choosing 5 from 36 ✓ correct number

- (b) only games are included.

(2 marks)

Solution
${}^{16}C_5 = 4\,368$
Specific behaviours
✓ indicates choosing 5 from 16 ✓ correct number

- (c) at least 3 games are included.

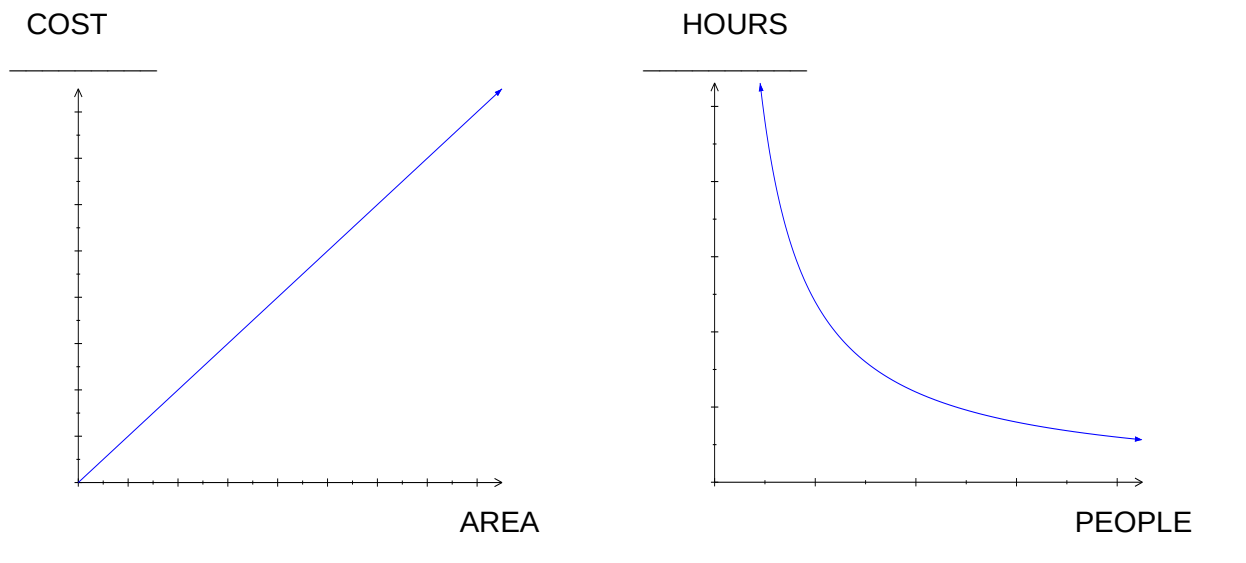
(3 marks)

Solution
${}^{16}C_3 {}^{20}C_2 + {}^{16}C_4 {}^{20}C_1 + {}^{16}C_5$ $= 106400 + 36400 + 4368$ $= 147168$
Specific behaviours
✓ choosing 3 games, 2 others ✓ includes 4 & 5 games ✓ correct answer

Question 11

(6 marks)

Trevor and Steven have decided to landscape their backyard including laying 80m^2 of artificial lawn at a cost of \$4000. They know that the two of them could complete the landscaping in 24 hours and that they have some friends who could help them if required.



- (a) One of the graphs shown depicts the total cost of artificial lawn for different areas and the other shows the total time taken for different numbers of people working on the landscaping.
Add the words **COST**, **AREA**, **HOURS** and **PEOPLE** to the axes above. (2 marks)

Specific behaviours
✓ direct labels correct
✓ inverse correct

- (b) Trevor and Steve decide to lay only 50m^2 of lawn. How much will this cost? (2 marks)

Solution
50×50 $= \$2500$
Specific behaviours
✓ direct relationship
✓ answer

- (c) Write an equation that relates the number of hours taken for the number of people landscaping. Hence, determine how long it would take six people to complete the landscaping. (2 marks)

Solution
$h = \frac{k}{p}$ $24 = \frac{k}{2}$ $k = 48$ $h = \frac{48}{p}$ $= \frac{48}{6}$ $= 8 \text{ hours}$
Specific behaviours
✓ correct equation for h
✓ correct hours

Question 12

(8 marks)

- (a) Determine the equation of the axis of symmetry for the graph of
- $y = 4x^2 - 20x - 51$
- .

(2 marks)

Solution
$x = -\frac{b}{2a} = -\frac{-20}{2 \times 4} = 2.5$ $x = 2.5$
Specific behaviours
✓ indicates use of formula ✓ correct equation

- (b) The graph of
- $y = ax^2 + bx + 27$
- passes through the points (3, 6) and (−4, −57). Determine the values of the constants
- a
- and
- b
- .

(3 marks)

Solution
$6 = (3)^2a + 3b + 27$ $6 = 9a + 3b + 27$ $-57 = (-4)^2a - 4b + 27$ $-57 = 16a - 4b + 27$ Solve simultaneously using CAS $a = -4, \quad b = 5$
Specific behaviours
✓ substitutes in both points ✓ solves for a ✓ solves for b

- (c) Determine the value(s) of
- m
- for which the graph of
- $y = x^2 - 4x + m$
- Has no real roots.

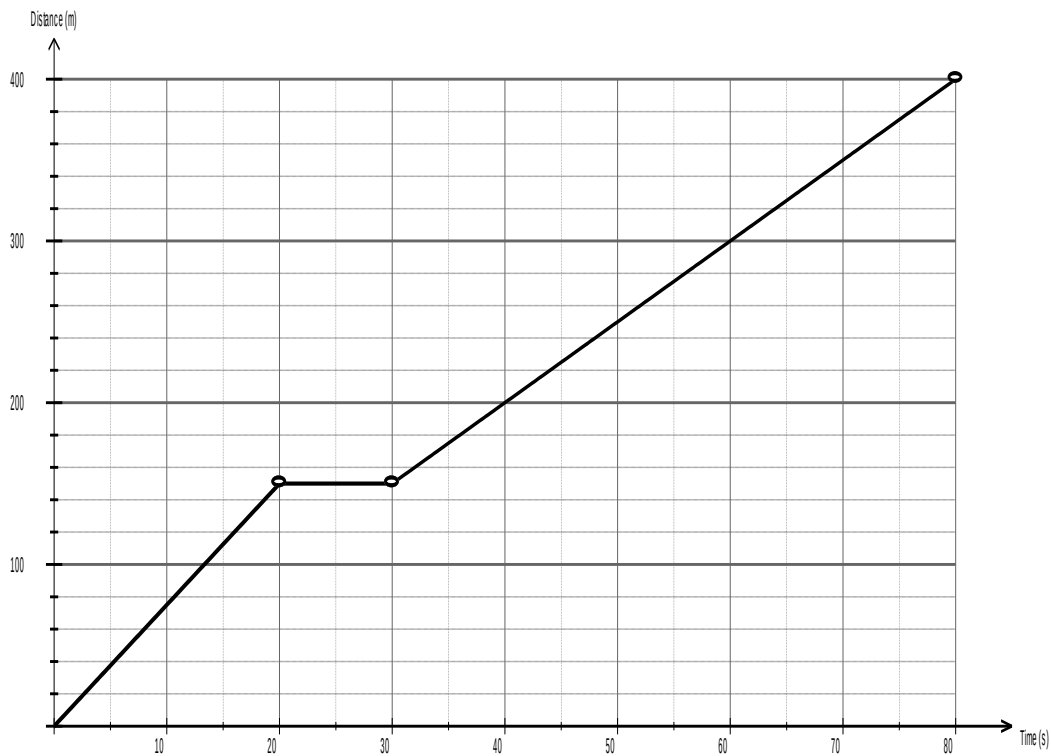
(3 marks)

Solution
$\Delta = 16 - 4 \times 1 \times m$ $\Delta < 0 \text{ when } m > 4$
Specific behaviours
✓ indicates use of Δ ✓ correct substitution ✓ $m > 4$

Question 13

(6 marks)

In the television special 'The World's Fittest Netball Player', Laura Geitz and Natalie von Bertouch take part in a 400m challenge. Laura's effort at the 400m course is shown below.



- (a) Determine the three different speeds in m/s travelled by Laura through the 400m.

(2 marks)

Solution
7.5m/s, 0m/s, 5m/s
Specific behaviours
✓ ✓ 3 correct
✓ 2 or 1

- (b) Natalie ran at 21.6km/h for the entire 400m. Determine her speed in m/s and draw her journey on the axes above.

Hence or otherwise determine when and where Natalie overtakes Laura.

(4 marks)

Solution
6m/s 25sec, 150m
Specific behaviours
✓ correct line for 6m/s
✓ correct start and finishing point of line
✓ 25sec
✓ 150m

Question 14**(7 marks)**

The current A (amperes), varies inversely to the resistance R (ohms) in an electric circuit.
When the resistance is 12 ohms, the current is 0.5 amperes.

- (a) State the constant of variation.

(1 mark)

Solution
$k = 6$
Specific behaviours
✓ k

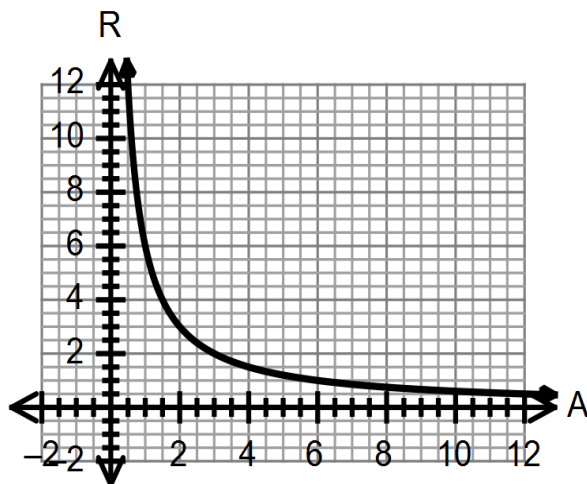
- (b) Calculate the resistance if the current is 5 amperes.

(1 mark)

Solution
$R = \frac{6}{5} = 1.2 \text{ ohms}$
Specific behaviours
✓ R

- (c) By first clearly plotting 2 points, draw the graph of this relationship on the axes below.
Label the axes.

(3 marks)



Specific behaviours
✓ plots 2 points correctly
✓ smooth curve
✓ axes labelled

- (d) Determine the effect on
- R
- if
- A
- is increased by 35%.

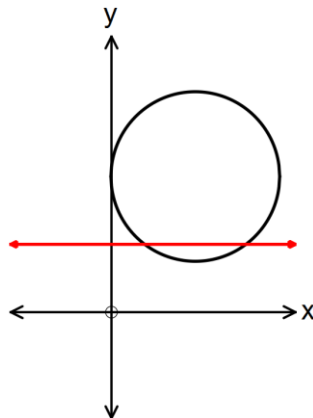
(2 marks)

Solution
$R = \frac{6}{A}$ $\left(\frac{1}{1.35}\right)R = \frac{6}{1.35A}$ R will decrease by 74%
Specific behaviours
✓ divides both sides by 1.35
✓ correct answer

Question 15

(8 marks)

The circle with centre A(5, 8) touches the y – axis as shown below.



- (a) By first expressing the equation of the circle in the form $(x - h)^2 + (y - k)^2 = r^2$ with centre (h, k) and radius r , show clearly that the equation of the circle is $x^2 + y^2 - 10x - 16y + 64 = 0$. (3 marks)

Solution
$(x - 5)^2 + (y - 8)^2 = 25$ $x^2 - 10x + 25 + y^2 - 16y + 64 = 25$ $x^2 + y^2 - 10x - 16y + 64 = 0$
Specific behaviours
✓ correct centre substitution ✓ correct radius ✓ shows correct working for expansion

- (b) The line $y = 4$ intersects the circle at points M and N.

- (i) Determine the coordinates of points M and N. (3 marks)

Solution
M(2, 4) N(8, 4)
Specific behaviours
✓ $x=2$ ✓ $x=8$ ✓ written as coordinates

- (ii) Determine the length of the chord MN. (2 marks)

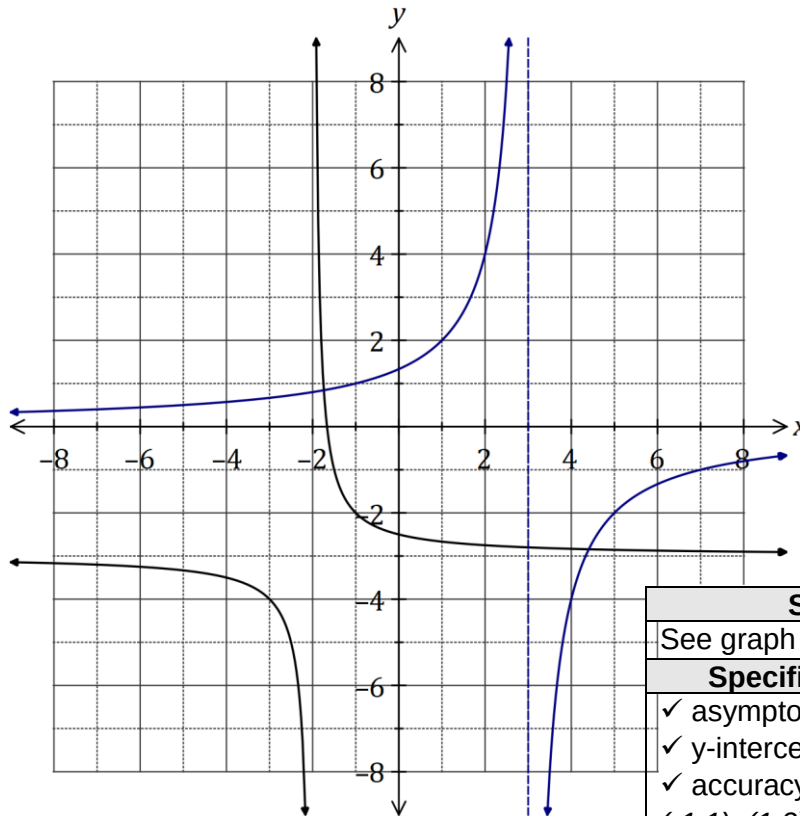
Solution
MN = 6 units
Specific behaviours
✓ correct subst into distance formula ✓ correct answer

Question 16

(8 marks)

Let $f(x) = \frac{4}{3-x}$ and $g(x) = \frac{1}{x+p} + q$, where p and q are constants.

The graph of $y = g(x)$ is shown below.



Solution
See graph
Specific behaviours
✓ asymptotes ✓ y-intercept ✓ accuracy [i.e. at least 3 of (-1,1), (1,2), (2,4), (4,-4) or (7,-1)]

- (a) Sketch the graph of $y = f(x)$ on the axes above.

(3 marks)

- (b) Determine the values of p and q .

(2 marks)

Solution
$p = 2, \quad q = -3$
Specific behaviours
✓ value of p ✓ value of q

- (c) Solve the equation $f(x) = g(x)$, giving your solution(s) to one decimal place.

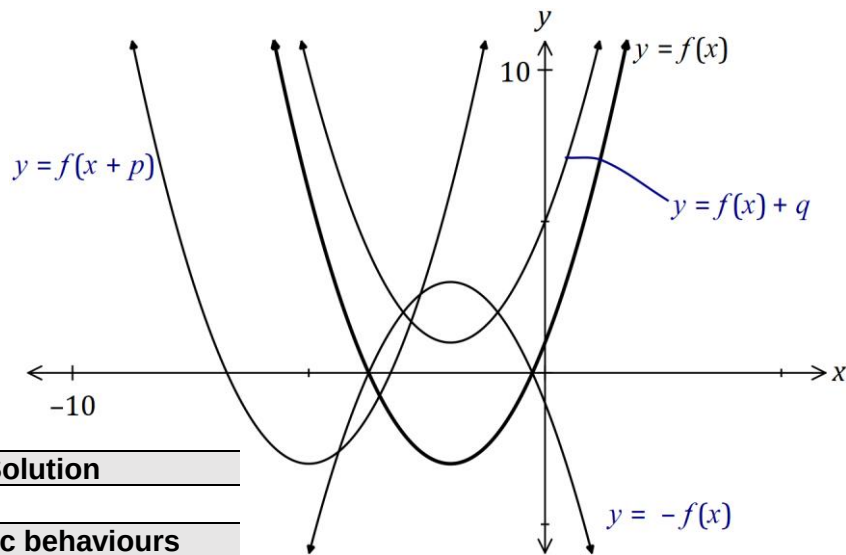
(3 marks)

Solution
$\frac{4}{3-x} = \frac{1}{x+2} - 3$ $x = -1.7, \quad x = 4.4$
Specific behaviours
✓ constructs equality ✓ one solution ✓ second solution (Rounding for guidance only but penalise answers given as coordinates -1 mark)

Question 17

(6 marks)

- (a) The graph of $y = f(x)$ is shown in bold below. The graphs of $y = -f(x)$, $y = f(x + p)$ and $y = f(x) + q$ are also shown, where p and q are constants.



Solution
See graph
Specific behaviours
✓✓✓ each correct label

Clearly label the remaining graphs with $y = -f(x)$, $y = f(x + p)$ or $y = f(x) + q$.

(3 marks)

- (b) The one-to-one relation $y = 7 - 3x$ has domain and range given by $\{x: x = -2, 3, a\}$ and $\{y: y = -8, -2, b\}$ respectively. Determine the values of constants a and b . (3 marks)

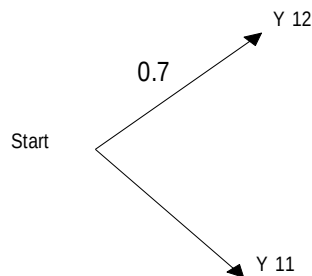
Solution
$x = 3, y = -2$
$x = -2, y = 13 = b$
$x = a, y = 7 - 3a = -8 \Rightarrow a = 5$
Specific behaviours
✓ value of b
✓ indicates a is mapped onto -8
✓ solves for value of a

Question 18

(9 marks)

In a group of 250 students, it is known that 70% are in Year 12 and the rest in Year 11. 40% of those in Year 12 and 80% of those in Year 11 study maths, while the remainder do not.

- (a) Complete the probability tree diagram to represent the above information.



Specific behaviours
✓✓ all correct -1 per omission

(2 marks)

- (b) Determine the probability that a randomly chosen student from the group

- (i) is in Year 11 and studies math.

(1 mark)

Solution
$P(\bar{T} \cap M) = 0.3 \times 0.8 = 0.24 \quad \left(\equiv \frac{6}{25} \right)$
Specific behaviours
✓ correct probability

- (ii) studies math.

(2 marks)

Solution
$P(T \cap M) + P(\bar{T} \cap M) = 0.7 \times 0.4 + 0.24 = 0.28 + 0.24 = 0.52 \quad \left(\equiv \frac{13}{25} \right)$
Specific behaviours
✓ indicates suitable method ✓ correct probability

- (iii) is in Year 11, given that they study math.

(2 marks)

Solution
$P(\bar{T} M) = \frac{0.24}{0.52} = \frac{6}{13} \quad (\approx 0.4615)$
Specific behaviours
✓ uses (i) and (ii) ✓ expresses P correctly (no need to fully simplify)

- (iv) is in Year 12 or studies math.

(2 marks)

Solution
$P(T) + P(M) - P(T \cap M) = 0.7 + 0.52 - 0.28 = 0.94 \quad \left(\equiv \frac{47}{50} \right)$
Specific behaviours
✓ indicates suitable method ✓ correct probability

Question 19

(8 marks)

For two events, X and Y , it is known that $P(X) = 0.48$ and $P(Y) = 0.35$. Determine the following probabilities, using any additional information only within that part of the question.

(a) $P(\bar{Y})$.

(1 mark)

Solution
$P(\bar{Y}) = 1 - 0.35 = 0.65$
Specific behaviours
✓ correct probability

(b) $P(Y \cap \bar{X})$ when $P(X \cap \bar{Y}) = 0.29$.

(2 marks)

Solution
$P(X \cap Y) = 0.48 - 0.29 = 0.19$
$P(Y \cap \bar{X}) = 0.35 - 0.19 = 0.16$
Specific behaviours
✓ calculates $P(X \cap Y)$
✓ correct probability

(c) $P(X|Y)$ when $P(Y|X) = 0.7$.

(3 marks)

Solution
$P(X \cap Y) = 0.48 \times 0.7 = 0.336$
$P(X Y) = \frac{0.336}{0.35} = 0.96$
Specific behaviours
✓ indicates use of conditional probability rule
✓ calculates $P(X \cap Y)$
✓ correct probability

(d) $P(X \cup Y)$ when X and Y are mutually exclusive.

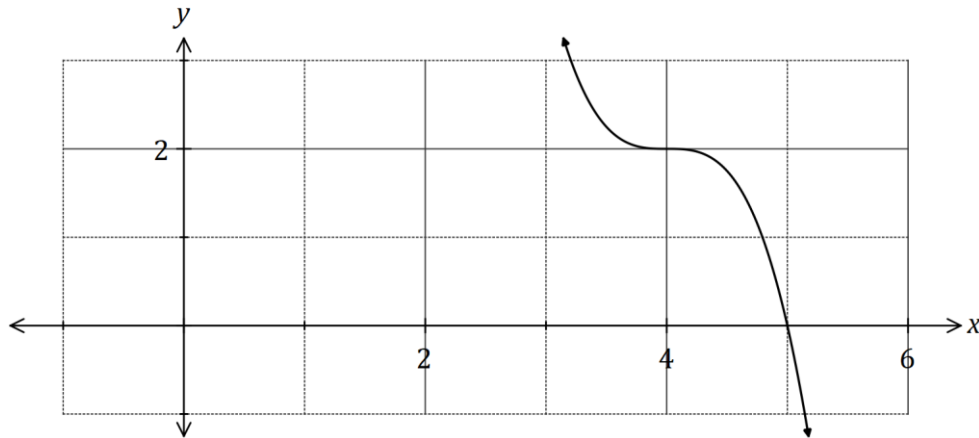
(2 marks)

Solution
$P(X \cap Y) = 0$
$P(X \cup Y) = 0.48 + 0.35$ $= 0.83$
Specific behaviours
✓ uses $P(X \cap Y) = 0$
✓ correct probability

Question 20

(6 marks)

- (a) Part of the graph of $y = f(x)$ is shown below, where $f(x) = -2(x - b)^3 + c$, and b and c are constants.



- (i) State the degree of $f(x)$.

Solution
3
Specific behaviours
✓ correct degree

(1 mark)

- (ii) Determine the value of b .

Solution
$b = 4$
Specific behaviours
✓ correct value

(1 mark)

- (iii) Determine $f(0)$.

Solution
$f(x) = -2(x - 4)^3 + 2$
$f(0) = -2(-4)^3 + 2 = 130$
Specific behaviours
✓ indicates value of c
✓ evaluates

(2 marks)

- (b) Another function is given by $g(x) = -f(x + 8)$.

Describe how to obtain the graph of $y = g(x)$ from the graph of $y = f(x)$.

(2 marks)

Solution
Translate graph 8 units to the left. Reflection about the x-axis
Specific behaviours
✓ correct translation
✓ correct reflection

Question 21

(9 marks)

Injury events in a state were grouped as follows: (D) injury events resulting in death; (H) injury events requiring hospitalisation; and (E) injury events requiring treatment at an emergency department. Injury events were further grouped according to whether prior use of alcohol was a contributing factor. The number of injury events for one month is summarised in the table below.

	D	H	E	Total
Alcohol	25	581	5 605	6 211
No Alcohol	115	4 422	11 912	16 449
Total	140	5 003	17 517	22 660

- (a) Calculate the percentage of all injury events that involved prior use of alcohol. (2 marks)

Solution
$\frac{6211}{22660} \times 100 = 27.4\%$
Specific behaviours
✓ identifies correct totals ✓ correct percentage

- (b) If one injury event was selected at random from the data, determine the probability that

- (i) it required hospitalisation.

Solution
$P(H) = \frac{5003}{22660} \approx 0.221$
Specific behaviours
✓ correct probability

(1 mark)

- (ii) it did not require treatment at an emergency department given that prior use of alcohol was a contributing factor. (2 marks)

Solution
$P(E A) = \frac{25 + 581}{6211} = \frac{606}{6211} \approx 0.0976$
Specific behaviours
✓ numerator ✓ denominator

- (iii) prior use of alcohol was a contributing factor given that it required hospitalisation. (2 marks)

Solution
$P(A H) = \frac{581}{5003} \approx 0.116$
Specific behaviours
✓ numerator ✓ denominator

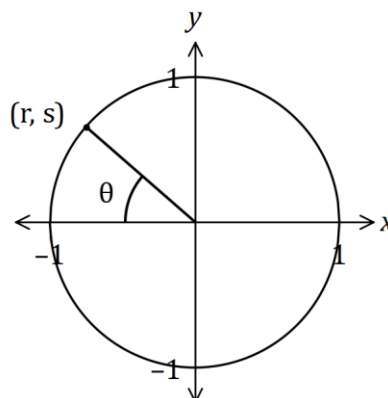
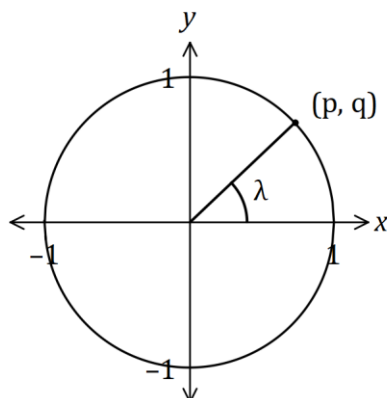
- (c) Use your answers above to show that alcohol and hospitalisation are not independent. (2 marks)

Solution
If independent $P(A H) = P(A)$ $0.116 \neq 0.274$ Therefore not independent.
Specific behaviours
✓ states independent relationship ✓ subst in probabilities

Question 22

(7 marks)

Consider the points with coordinates (p, q) and (r, s) that lie in the first and second quadrants respectively of the unit circles shown below, where λ and θ are acute angles.



Determine the following in terms of p, q, r and s , simplifying your answers where possible.

(a) $\sin \theta$.

(1 mark)

(b) $\tan \theta$.

(2 marks)

(c) $\cos(\pi + \lambda)$.

(2 marks)

(d) $\sin(90^\circ - \lambda)$.

(2 marks)

Solutions	
(a) s ✓	
(b) $\frac{s}{r}$ ✓	
(c) $-p$ ✓✓ (-1 if p)	
(d) p ✓✓ (-1 if $-p$)	
Specific behaviours	
✓ each correct response	

Supplementary page

Question number: _____

Supplementary page

Question number: _____

